

Final

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Problem 1

The priors for β_{0rs} and β_{1rs} can be considered noninformative because they are normally distributed with very small precision which mimics a uniform distribution as the precision approaches 0.

The model converged reasonably well. After 5000 iterations of burn in, the history plots show no need for more burn in and resemble white noise, the density plots are somewhat bumpy but not concerningly so, and the ACF shows some autocorrelation for the first few lags but not much after that.

The sex-specific ratio between black and white death rates can be tracked by adding the following code to the model

```
for (s in 1:2) {  
  for (t in 1:10) {  
    log_ratio[s, t] <- (beta0[1, s] - beta0[2, s]) + (beta1[1, s] - beta1[2, s]) * (t - 1)  
    ratio_bw[s, t] <- exp(log_ratio[s, t])  
  }  
}
```

The mean of the black/white death rate ratio for each sex in each year is shown below:

We can see that the ratio is smaller for men than it is for women and that it increased over time during the study period for both men and women.

There are eight parameters in this model because there are four different combinations of race and sex and there are two parameters (β_0 and β_1) for each combination. This is very similar to the p_D estimate in WinBUGS of 8.054.

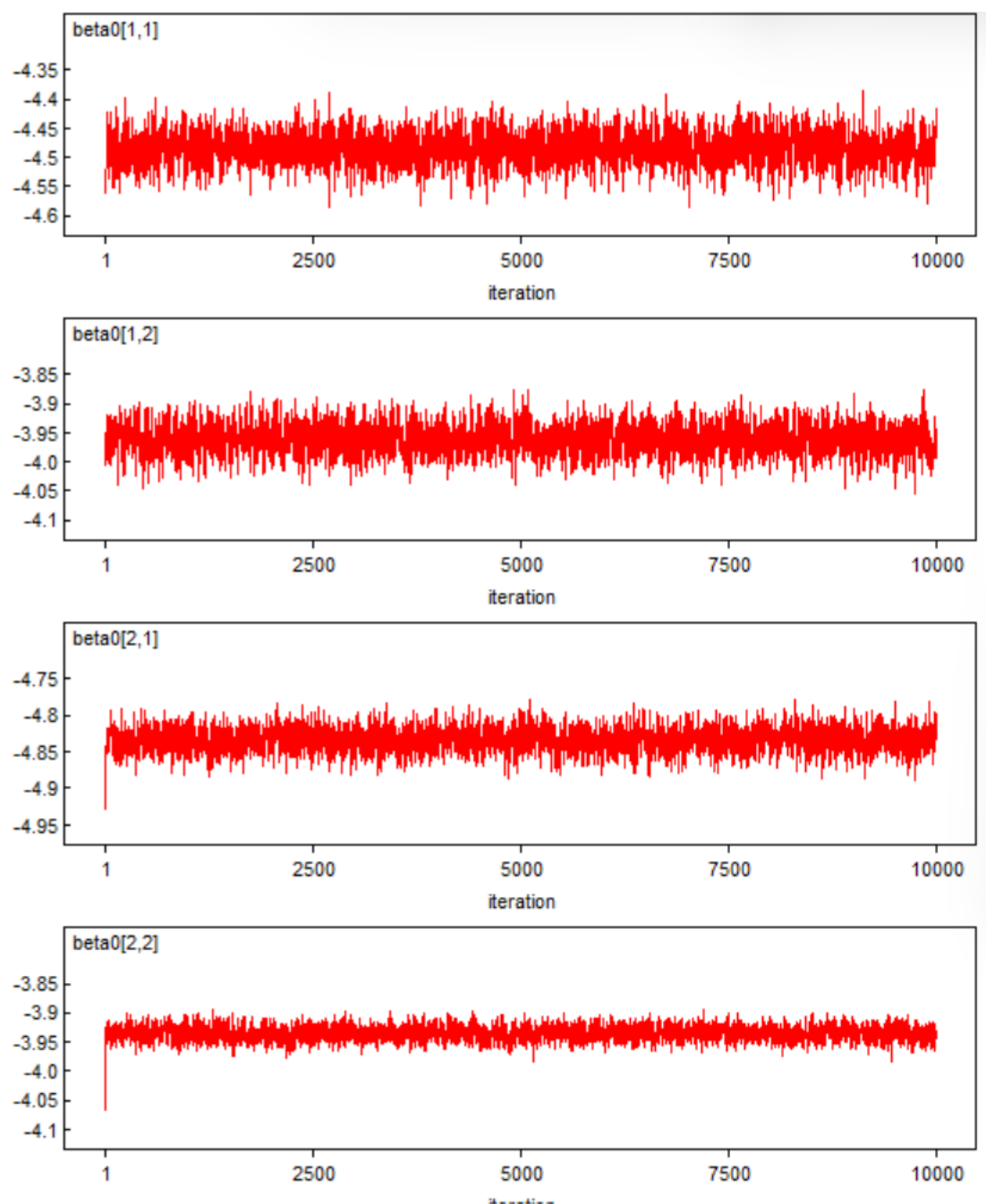


Figure 1: Beta 0 History Plots

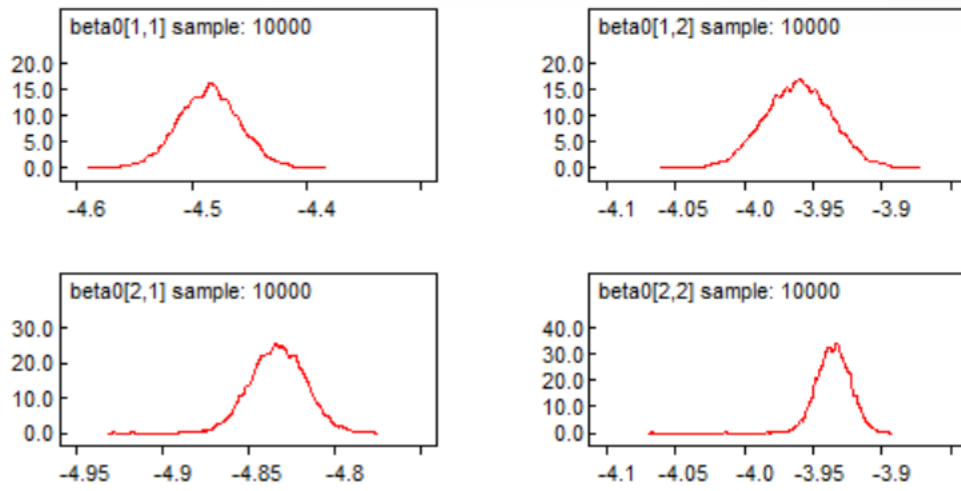


Figure 2: Beta 0 Density Plots

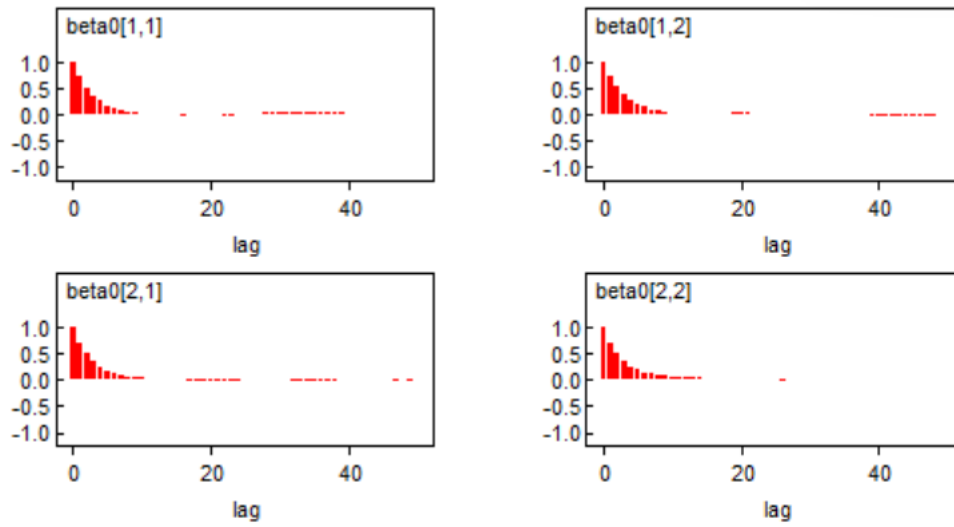


Figure 3: Beta 0 ACF Plots

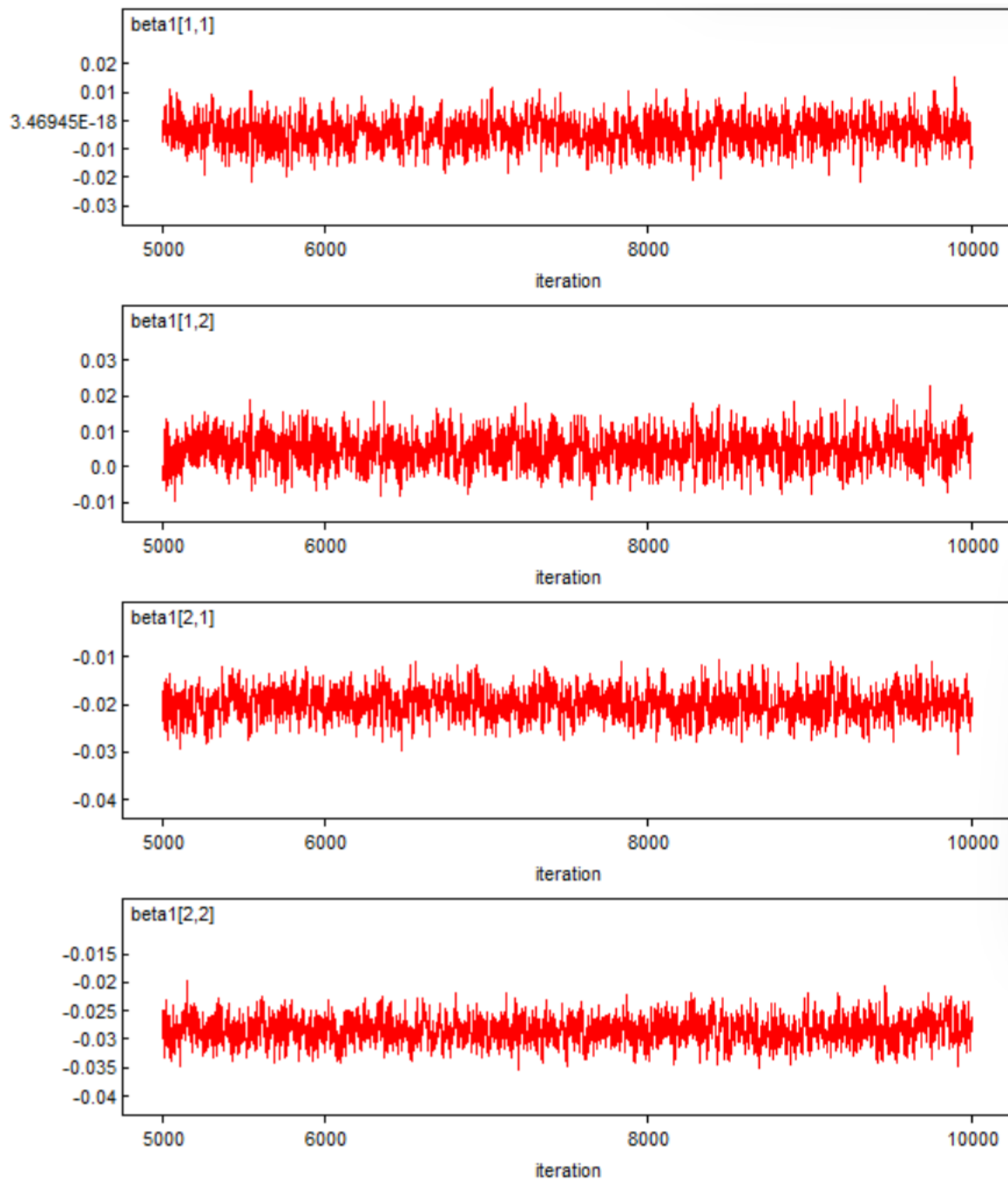


Figure 4: Beta 1 History Plots

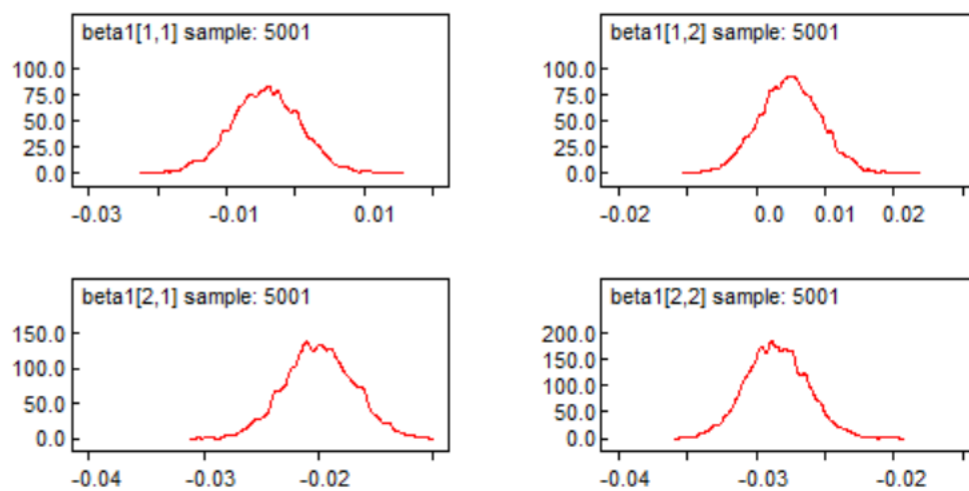


Figure 5: Beta 1 Density Plots

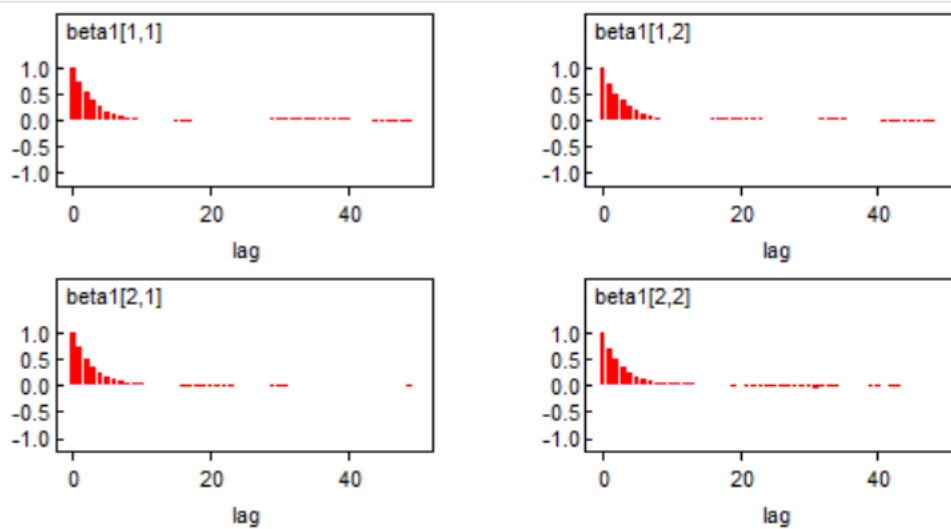


Figure 6: Beta 1 ACF Plots

node	mean
ratio_bw[1,1]	1.418
ratio_bw[1,2]	1.44
ratio_bw[1,3]	1.462
ratio_bw[1,4]	1.485
ratio_bw[1,5]	1.508
ratio_bw[1,6]	1.532
ratio_bw[1,7]	1.556
ratio_bw[1,8]	1.581
ratio_bw[1,9]	1.606
ratio_bw[1,10]	1.631
ratio_bw[2,1]	0.9736
ratio_bw[2,2]	1.007
ratio_bw[2,3]	1.041
ratio_bw[2,4]	1.076
ratio_bw[2,5]	1.113
ratio_bw[2,6]	1.15
ratio_bw[2,7]	1.189
ratio_bw[2,8]	1.23
ratio_bw[2,9]	1.272
ratio_bw[2,10]	1.315

Figure 7: Sex-specific Black/White Death Rate Ratios

	Dbar = post.mean of -2logL; Dhat = -2LogL at post.mean of stochastic nodes			
	Dbar	Dhat	pD	DIC
Y	19004.500	18996.400	8.054	19012.500
total	19004.500	18996.400	8.054	19012.500

Figure 8: WinBUGS DIC Estimate

Problem 2

The following code was used to create the model for this problem:

```
model{
  for (i in 1:N) {
    Y[i] ~ dpois(nlambda[i])
    nlambda[i] <- n[i] * lambda[i]

    eta[i] <- beta0[race[i], sex[i]] +
      beta1[race[i], sex[i]] * (year[i] - 1) +
      z[county[i], race[i], sex[i]] +
      w[county[i], race[i], sex[i]] * (year[i] - 1)

    log_lambda[i] ~ dnorm(eta[i], tau0[race[i], sex[i]])
    lambda[i] <- exp(log_lambda[i])
  }
  for(r in 1:Nr){
    for(s in 1:Ng){
      beta0[r,s] ~ dnorm(0,0.001)
      beta1[r,s] ~ dnorm(0,0.001)
      tau0[r, s] ~ dgamma(1,0.01)
      tau.z[r, s] ~ dgamma(1,0.147)
      tau.w[r, s] ~ dgamma(1,0.147)
    }
  }

  # r = 1, s = 1
  z_group11[1:Ns] ~ car.normal(adj[], weights[], num[], tau.z[1,1])
  w_group11[1:Ns] ~ car.normal(adj[], weights[], num[], tau.w[1,1])
  for (i in 1:Ns) {
    z[i,1,1] <- z_group11[i]
    w[i,1,1] <- w_group11[i]
  }

  # r = 1, s = 2
  z_group12[1:Ns] ~ car.normal(adj[], weights[], num[], tau.z[1,2])
  w_group12[1:Ns] ~ car.normal(adj[], weights[], num[], tau.w[1,2])
  for (i in 1:Ns) {
    z[i,1,2] <- z_group12[i]
    w[i,1,2] <- w_group12[i]
  }

  # r = 2, s = 1
  z_group21[1:Ns] ~ car.normal(adj[], weights[], num[], tau.z[2,1])
  w_group21[1:Ns] ~ car.normal(adj[], weights[], num[], tau.w[2,1])
  for (i in 1:Ns) {
    z[i,2,1] <- z_group21[i]
    w[i,2,1] <- w_group21[i]
  }

  # r = 2, s = 2
  z_group22[1:Ns] ~ car.normal(adj[], weights[], num[], tau.z[2,2])
  w_group22[1:Ns] ~ car.normal(adj[], weights[], num[], tau.w[2,2])
  for (i in 1:Ns) {
```

```

z[i,2,2] <- z_group22[i]
w[i,2,2] <- w_group22[i]
}

for (s in 1:2) {
  for (t in 1:10) {
    log_ratio[s, t] <- (beta0[1, s] - beta0[2, s]) + (beta1[1, s] - beta1[2, s]) * (t - 1)
    ratio_bw[s, t] <- exp(log_ratio[s, t])
  }
}

for (k in 1:sumNumNeigh) {
  weights[k] <- 1
}
junk <- county[1] + Ns + Ny + sumNumNeigh
}

```

This model did not converge as well as the model in problem 1. The history and ACF plots both show significant autocorrelation and the density plots look much less normal, especially β_1 for white women.

The DIC of this model is slightly lower than that of the previous model. The number of parameters in this model is 820 because there are 400 zs (one for each combo of county, sex, and race), 400 ws (same as z), four β_0 s and β_0 s as before, and four of each of the three tau parameters because there is one for each sex-race combo. This is higher than the 808.983 estimated by WinBUGS. They would be much closer if you didn't include the τ parameters, suggesting that these may not have much effect on the deviance of the model.

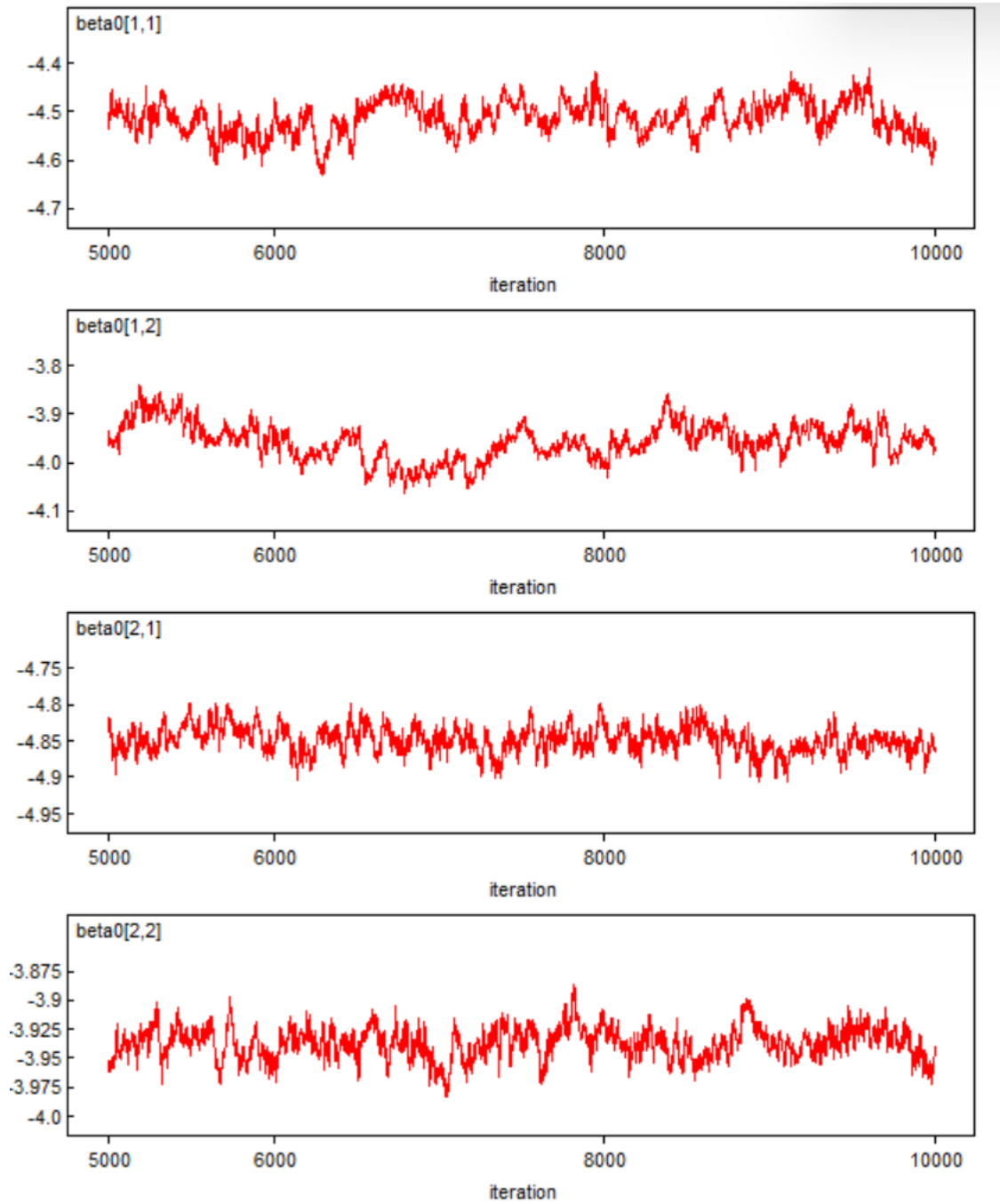


Figure 9: Beta 0 History Plots

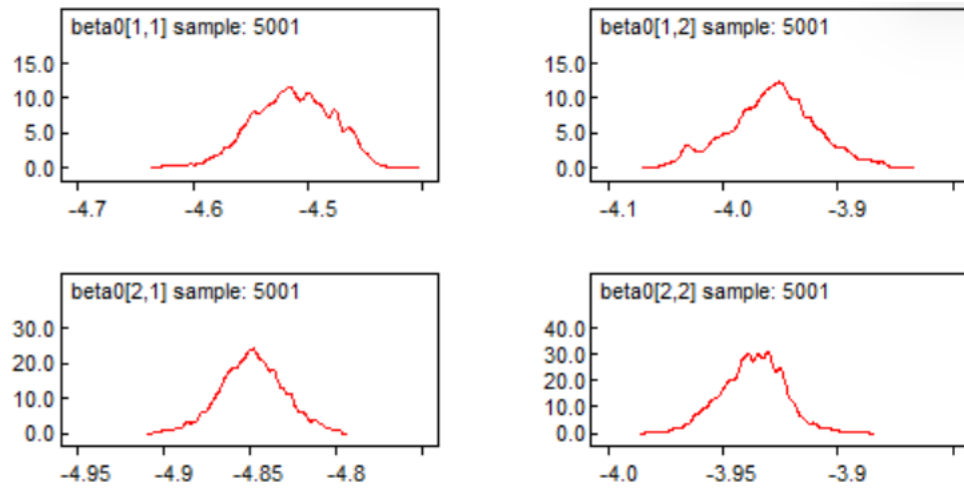


Figure 10: Beta 0 Density Plots

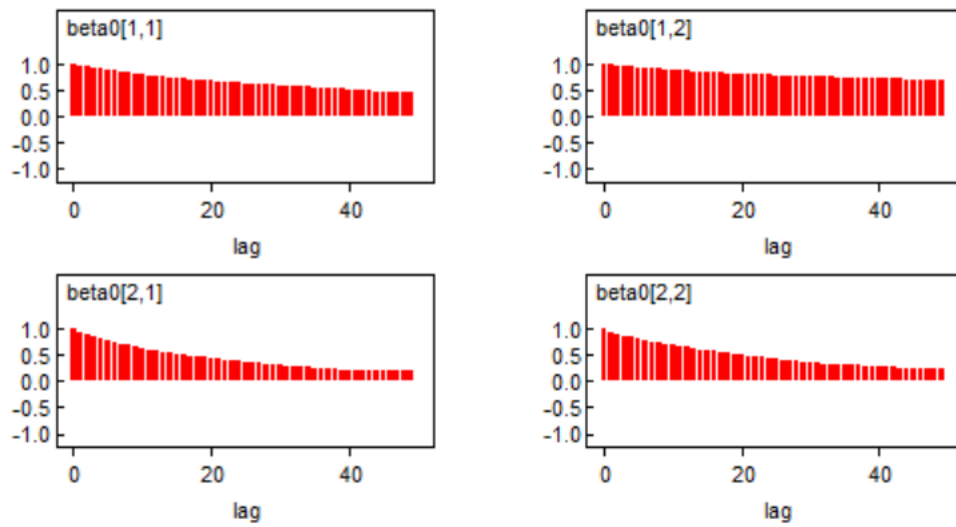


Figure 11: Beta 0 ACF Plots

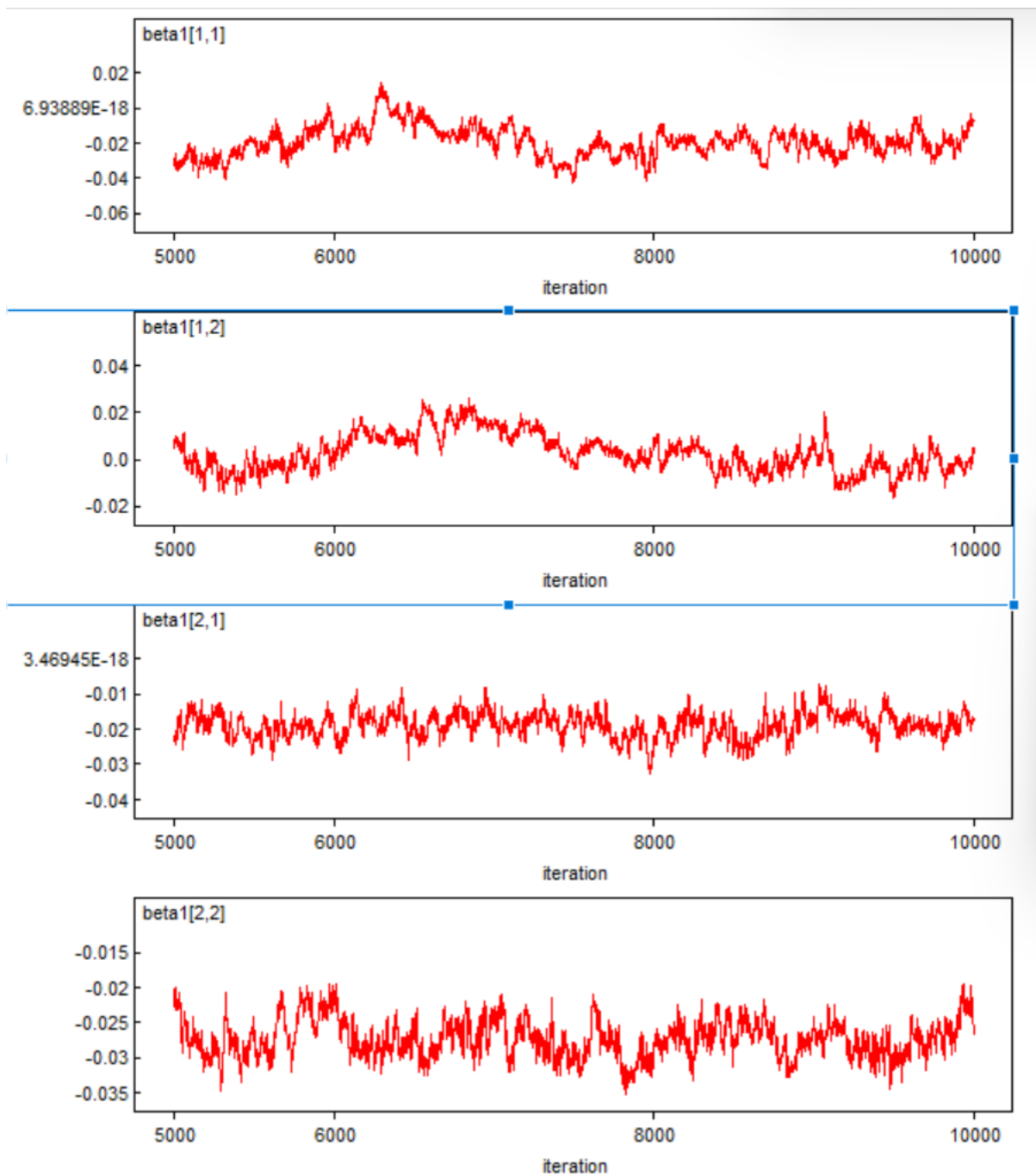


Figure 12: Beta 1 History Plots

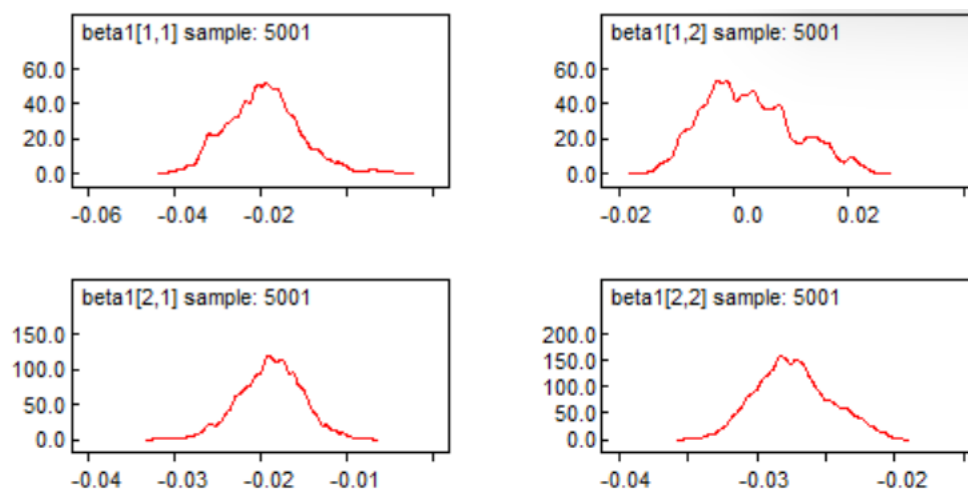


Figure 13: Beta 1 Density Plots

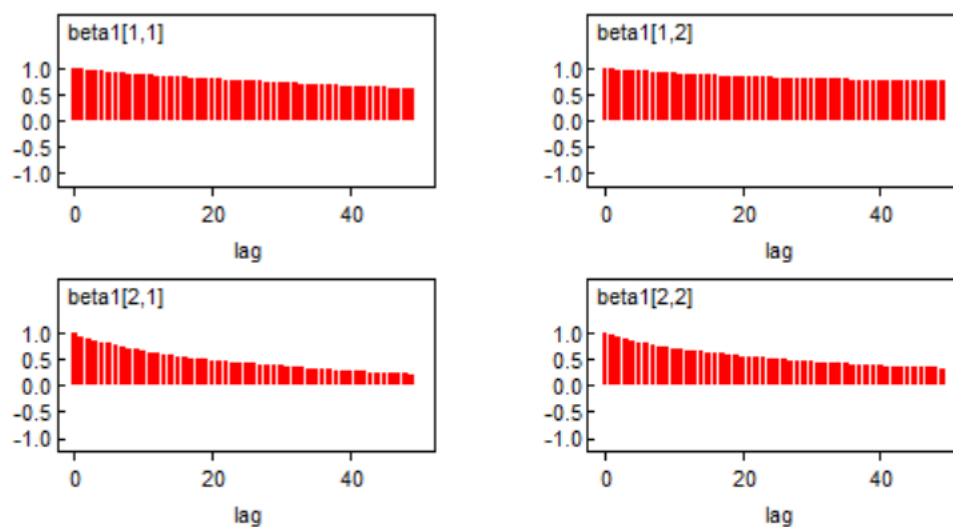


Figure 14: Beta 1 ACF Plots

Dbar = post.mean of -2logL; Dhat = -2LogL at post.mean of stochastic nodes

	Dbar	Dhat	pD	DIC
Y	18168.500	17359.500	808.983	18977.500
total	18168.500	17359.500	808.983	18977.500

Figure 15: WinBUGS DIC Estimate

Problem 3

For this MCAR model, the τ_z and τ_w parameters from the model in the previous problem would be replaced by 4x4 covariance matrices, with only one for each race-sex combo. Each of these matrices are symmetric so they effectively only have 10 free parameters, the diagonal and one side of the diagonal, rather than 16. Since Everything else remains the same as for the model in the previous problem, the model has 832 parameters.

Examining the posterior of these covariance matrices can be helpful in diagnosing whether the MCAR structure was beneficial. If there are high values outside of the diagonal, this reflects that there is significant correlation across groups so the MCAR structure is useful.

Problem 4

```
library(coda)
```

```
## Warning: package 'coda' was built under R version 4.4.3
```

```
samples <- read.coda(  
  output.file = "lambdas_2.txt",  
  index.file = "index.txt"  
)
```

```
lambda_array <- array(samples, dim = c(500, 10, 2, 2, 100))  
lambda_t1 = lambda_array[,1,,,]  
lambda_t10 = lambda_array[,10,,,]  
#Medians not means, I just didn't want to change the names throughout the code  
lambda_means_1 <- apply(lambda_t1, c(2, 3, 4), median)  
lambda_means_10 <- apply(lambda_t10, c(2, 3, 4), median)
```

Get NC map

```
library(maps)
```

```
## Warning: package 'maps' was built under R version 4.4.3
```

```
load('Code/County Shapefiles/nc.rdata')
```

Map code for 1979

```
library(RColorBrewer)
```

```
ncols <- 7
```

```
for (s in 1:2) {  
  for (r in 1:2) {  
    cols=brewer.pal(ncols, 'RdYlBu')[ncols:1]  
    tcuts=quantile(lambda_means_1[s, r, ]*100000,1:(ncols-1)/ncols)  
    tcolb=array(rep(lambda_means_1[s, r, ]*100000,each=ncols-1) > tcuts,  
      dim=c(ncols-1,100))  
    tcol =apply(tcolb,2,sum)+1  
  
    filename <- paste0("lambda_r", as.numeric(r), "_s", as.numeric(s), "_t", 1, ".png")  
  
    png(filename, width = 900, height = 300)  
    plot(nc, col=cols[tcol], border = "grey",  
      main = paste("Race", as.numeric(r), "- Sex", as.numeric(s), "- Year", 1))  
    legend('right',inset=c(-.01,0),xpd=TRUE,  
      legend=c(paste(  
        c('Below',round(tcuts[-(ncols-1)],0),'Over'),  
        c(' ',rep(' - ',ncols-2),' '),  
        c(round(tcuts,0),round(tcuts[ncols-1],0)),sep='')),  
      fill=cols,title='Deaths per 100,000',bty='n',cex=1.5,  
      border='lightgray')  
    dev.off()  
  }  
}
```

Map code for 1988

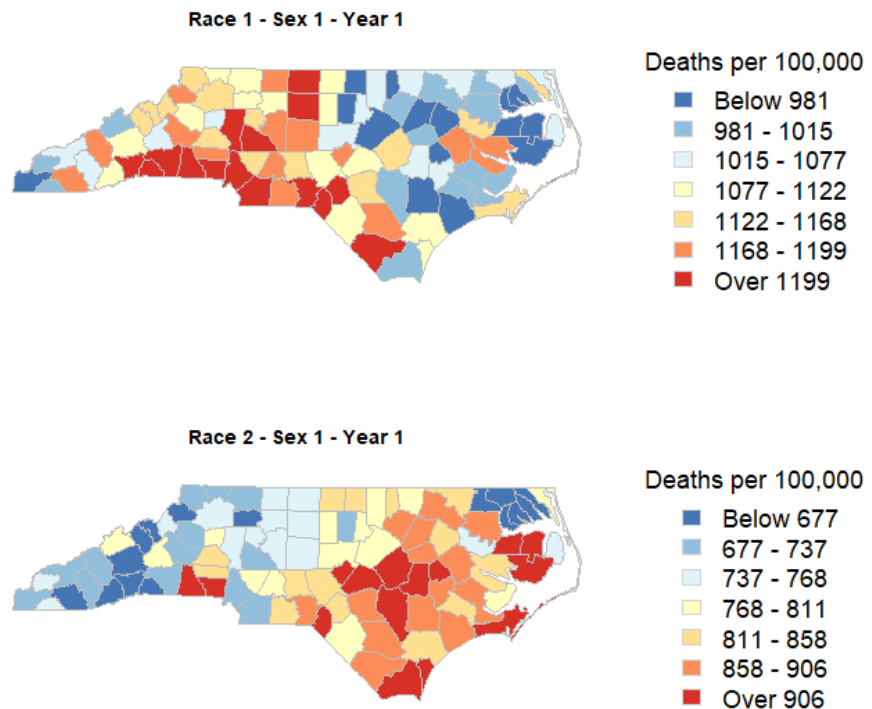
```

for (s in 1:2) {
  for (r in 1:2) {
    cols=brewer.pal(ncols,'RdYlBu')[ncols:1]
    tcuts=quantile(lambda_means_10[s, r, ]*100000,1:(ncols-1)/ncols)
    tcolb=array(rep(lambda_means_10[s, r, ]*100000,each=ncols-1) > tcuts,
dim=c(ncols-1,100))
    tcol =apply(tcolb,2,sum)+1

    filename <- paste0("lambda_r", as.numeric(r), "_s", as.numeric(s), "_t", 10, ".png")

    png(filename, width = 900, height = 300)
    plot(nc, col=cols[tcol], border = "grey",
        main = paste("Race", as.numeric(r), "- Sex", as.numeric(s), "- Year", 10))
    legend('right',inset=c(-.01,0),xpd=TRUE,
        legend=c(paste(
c('Below',round(tcuts[-(ncols-1)],0),'Over'),
c(' ',rep(' - ',ncols-2),' '),
c(round(tcuts,0),round(tcuts[ncols-1],0)),sep='')),
        fill=cols,title='Deaths per 100,000',bty='n',cex=1.5,
        border='lightgray')
    dev.off()
  }
}

```



The race/sex combinations exhibit some common geographic trends in both years, but it is fairly limited. White men and women have counties with high rates in a very similar geographic region, but this differs from those of Black men and women and those of Black men and women differ from each other.

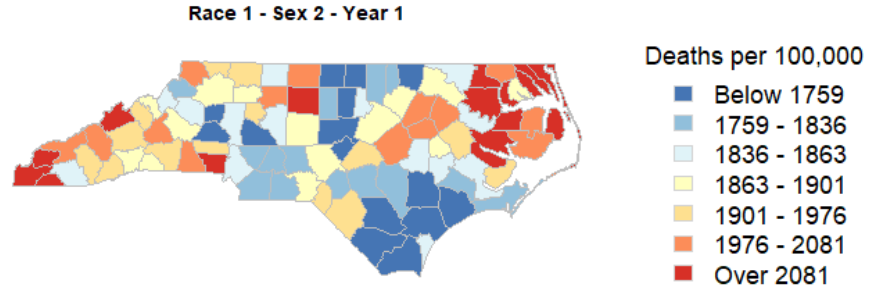


Figure 16: Median Heart Disease Death Rate Per 100,000 for Black Men in 1979

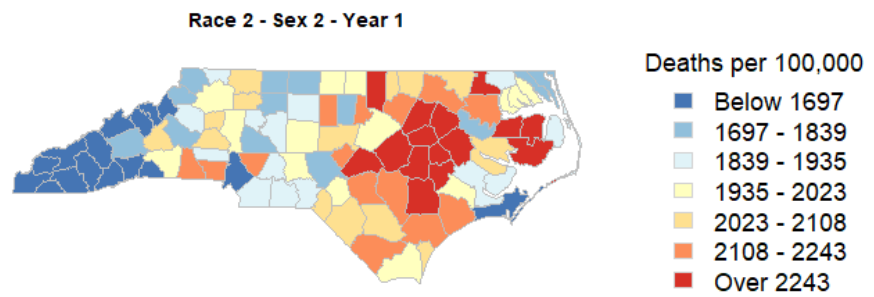


Figure 17: Median Heart Disease Death Rate Per 100,000 for White Men in 1979

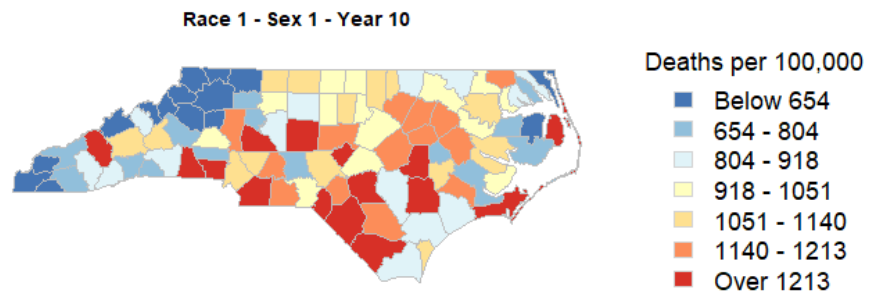


Figure 18: Median Heart Disease Death Rate Per 100,000 for Black Women in 1988

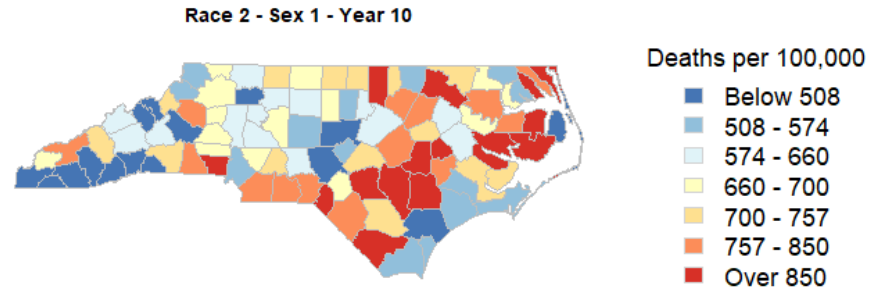


Figure 19: Median Heart Disease Death Rate Per 100,000 for White Women in 1988

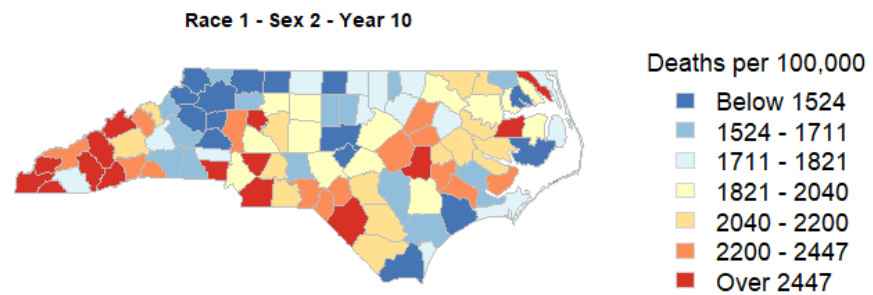


Figure 20: Median Heart Disease Death Rate Per 100,000 for Black Men in 1988

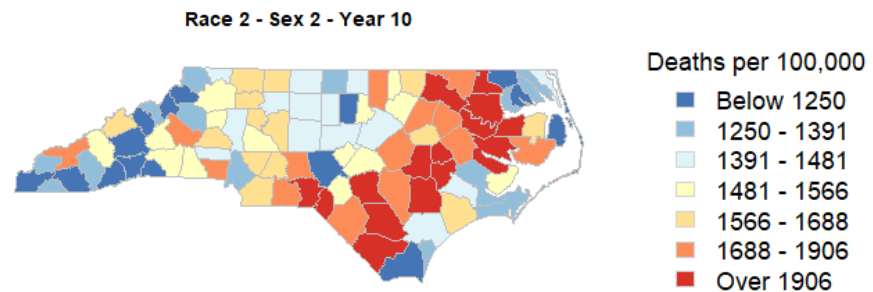


Figure 21: Median Heart Disease Death Rate Per 100,000 for White Men in 1988

Ratio map code for 1979

```
ncols <- 7

for (s in 1:2) {
  ratio_vals <- lambda_means_1[s, 1, ] / lambda_means_1[s, 2, ]

  cols <- brewer.pal(ncols, 'RdYlBu')[ncols:1]
  tcuts <- quantile(ratio_vals, probs = 1:(ncols - 1) / ncols, na.rm = TRUE)
  tcolb <- array(rep(ratio_vals, each = ncols - 1) > tcuts,
                 dim = c(ncols - 1, length(ratio_vals)))
  tcol <- apply(tcolb, 2, sum) + 1

  filename <- paste0("lambda_ratio_bw_s", s, "_t1.png")

  png(filename, width = 900, height = 300)
  plot(nc, col = cols[tcol], border = "grey",
       main = paste("Black / White Death Rate Ratio - Sex", s, "- Year 1"))
  legend('right', inset = c(-.01, 0), xpd = TRUE,
        legend = paste(
          c('Below', round(tcuts[-(ncols - 1)], 2), 'Over'),
          c(' ', rep(' - ', ncols - 2), ' '),
          c(round(tcuts, 2), round(tcuts[ncols - 1], 2)), sep = ''),
        fill = cols, title = 'Lambda Ratio (Black/White)',
        bty = 'n', cex = 1.5, border = 'lightgray')
  dev.off()
}
```

Ratio map code for 1988

```
ncols <- 7

for (s in 1:2) {
  ratio_vals <- lambda_means_10[s, 1, ] / lambda_means_10[s, 2, ]

  cols <- brewer.pal(ncols, 'RdYlBu')[ncols:1]
  tcuts <- quantile(ratio_vals, probs = 1:(ncols - 1) / ncols, na.rm = TRUE)
  tcolb <- array(rep(ratio_vals, each = ncols - 1) > tcuts,
                 dim = c(ncols - 1, length(ratio_vals)))
  tcol <- apply(tcolb, 2, sum) + 1

  filename <- paste0("lambda_ratio_bw_s", s, "_t10.png")

  png(filename, width = 900, height = 300)
  plot(nc, col = cols[tcol], border = "grey",
       main = paste("Black / White Death Rate Ratio - Sex", s, "- Year 10"))
  legend('right', inset = c(-.01, 0), xpd = TRUE,
        legend = paste(
          c('Below', round(tcuts[-(ncols - 1)], 2), 'Over'),
          c(' ', rep(' - ', ncols - 2), ' '),
          c(round(tcuts, 2), round(tcuts[ncols - 1], 2)), sep = ''),
        fill = cols, title = 'Lambda Ratio (Black/White)',
        bty = 'n', cex = 1.5, border = 'lightgray')
  dev.off()
}
```

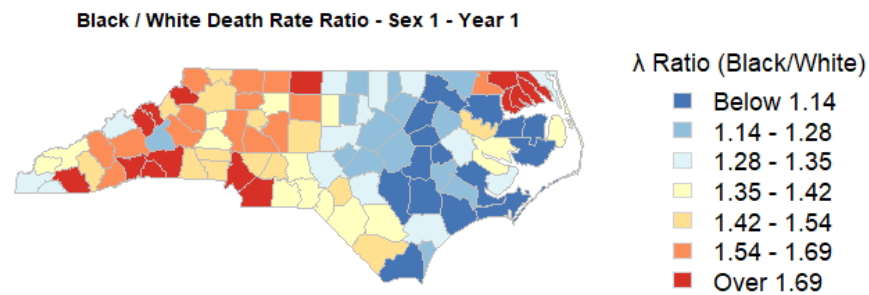


Figure 22: Black/White Ratio of Heart Disease Death Rates for Women in 1979

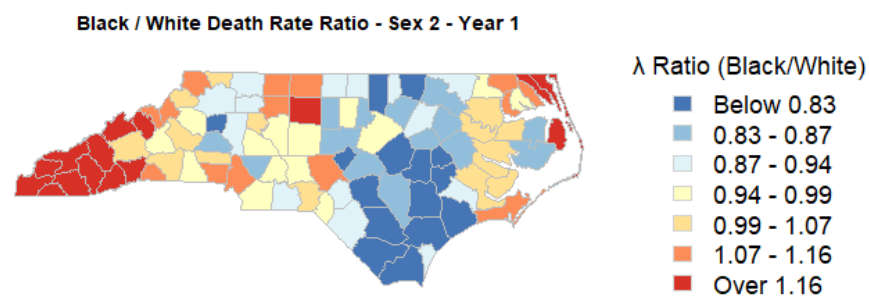


Figure 23: Black/White Ratio of Heart Disease Death Rates for Men in 1979

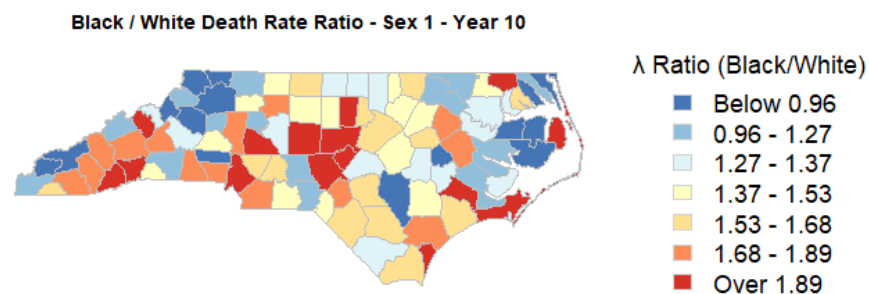


Figure 24: Black/White Ratio of Heart Disease Death Rates for Women in 1988

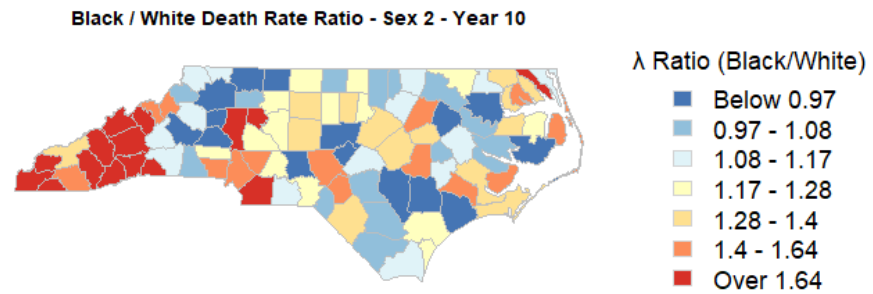


Figure 25: Black/White Ratio of Heart Disease Death Rates for Men in 1988

There is significant clustering in these ratios. The western half of the state has much higher ratios than the eastern half. This is more pronounced for men than for women. This pattern didn't change much between 1979 and 1988 but it appears to have become slightly less significant for women.

Problem 5

Models that could be used to capture more complex trends over a longer study period include piecewise linear models, which could be useful if there are sharp changes in the trend, and polynomial regression models, which could be useful if the trend is smooth but nonlinear. The piecewise linear model would use the same priors and parameters for each piece, while the polynomial regression model would include additional β parameters with noninformative normal priors similar to the existing β s.